2) Apache SystemML: An optimal workplace for machine learning using Big Data

SystemML is the machine learning technology created at IBM. It ranks among the top-level projects at the Apache Software Foundation. It's a flexible, scalable machine learning system.

Apache SystemML™

Important characteristics

1. Algorithm customisability via R-like and Python-like languages
2. Multiple execution modes, including Spark MLContext, Spark Batch, Hadoop Batch, Standalone and JMLC (Java Machine Learning Connector)
3. Automatic optimisation based on data and cluster characteristics to ensure both efficiency and scalability

SystemML is considered as the SQL for machine learning. The latest version (1.0.0) of SystemML supports Java 8+, Scala 2.11+, Python 2.7/3.5+, Hadoop 2.6+ and Spark 2.1+.

It can be run on top of Apache Spark, where it automatically scales your data, line by line, determining whether your code should be run on the driver or an Apache Spark cluster. Future SystemML developments include additional deep learning with GPU capabilities, such as importing and running neural network architectures and pre-trained models for training.

Java Machine Learning Connector (JMLC) for SystemML

The Java Machine Learning Connector (JMLC) API is a programmatic interface for interacting with SystemML in an embedded fashion. The primary purpose of JMLC is that of a scoring API, whereby your scoring function is expressed using SystemML's DML (Declarative Machine Learning) language. In addition to scoring, embedded SystemML can be used for tasks such as unsupervised learning (like clustering) in the context of a larger application running on a single machine.

Useful links

SystemML home: *https://systemml.apache.org/*
GitHub: *https://github.com/apache/systemml*

3) Caffe: A deep learning framework made with expression, speed and modularity in mind

The Caffe project was initiated by Yangqing Jia during the course of his Ph.D at UC Berkeley, and later developed further by Berkeley AI Research (BAIR) and community contributors. It mostly focuses on convolutional networks for computer vision applications. Caffe is a solid, popular choice for computer vision-related tasks, and you can download many successful models made by Caffe users from the Caffe Model Zoo (link below) for out-of-the-box use.

Caffe's advantages

1) Expressive architecture encourages application and innovation. Models and optimisation are defined by configuration without hard coding. Users can switch between CPU and GPU by setting a single flag to train on a GPU machine, and then deploy to commodity clusters or mobile devices.
2) Extensible code fosters active development. In Caffe's first year, it was forked by over 1,000 developers and had many significant changes contributed back.
3) Speed makes Caffe perfect for research experiments and industry deployment. Caffe can process over 60 million images per day with a single NVIDIA K40 GPU.
4) *Community:* Caffe already powers academic research projects, startup prototypes, and even large-scale industrial applications in vision, speech and multimedia.

Useful links

Caffe home: *http://caffe.berkeleyvision.org/*
GitHub: *https://github.com/BVLC/caffe*
Caffe user group: *https://groups.google.com/forum/#!forum/caffe-users*

Tutorial presentation of the framework and a full-day crash course: *https://docs.google.com/presentation/d/1UeKXVgRvvxg9OUdh_UiC5G71UMscNPlvArsWER41PsU/edit#slide=id.p*

Caffe Model Zoo: *https://github.com/BVLC/caffe/wiki/Model-Zoo*

4) Apache Mahout: A distributed linear algebra framework and mathematically expressive Scala DSL

Mahout is designed to let mathematicians, statisticians and data scientists quickly implement their own algorithms. Apache Spark is the recommended out-of-the-box distributed back-end or can be extended to other distributed back-ends. Its features include the following:

- It is a mathematically expressive Scala DSL
- Offers support for multiple distributed back-ends (including Apache Spark)
- Has modular native solvers for CPU, GPU and CUDA acceleration

Apache Mahout currently implements collaborative filtering (CF), clustering and categorisation.

Features and applications

- Taste CF: Taste is an open source project for CF (collaborative filtering) started by Sean Owen on SourceForge and donated to Mahout in 2008
- Several Map-Reduce enabled clustering implementations, including k-Means, fuzzy k-Means, Canopy, Dirichlet and Mean-Shift
- Distributed Naive Bayes and Complementary Naive Bayes